Sameer Joshi

Embedded Firmware Engineer

RESUME_1860

Location: Raipur, Chhattisgarh, India | Email: sameer.joshi.resume_1860@example.com | Phone: +91-00000-00000 (placeholder)
Total Experience: 15 years

PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 15 years of experience in embedded firmware, using C, C++, ARM Cortex-M, FreeRTOS. Experienced in real-time systems, hardware debugging and measurable delivery outcomes.

TECHNICAL SKILLS

Skills: STM32, JTAG, ARM Cortex-M, UART, C, Oscilloscope, Firmware Debugging, FreeRTOS, C++, Git, Embedded Linux, SPI

Methods: Real-time systems, Hardware debugging, Protocol validation

PROFESSIONAL EXPERIENCE

Lead Embedded Firmware Engineer | Ironleaf Engineering | Jan 2021 - Present

- Lowered field-reset incidents by 37% with watchdog recovery and diagnostics
- Validated 311 hardware test vectors across protocol and timing corner cases
- Improved battery runtime by 41% with peripheral and sleep-state optimization
- Coordinated engineering design reviews, acceptance criteria and test evidence with cross-functional stakeholders.

Embedded Firmware Engineer | HelioNex Labs | Jan 2016 - Dec 2020

- Improved battery runtime by 38% with peripheral and sleep-state optimization
- Reduced embedded task jitter by 17% using interrupt priority and queue tuning

Associate Embedded Firmware Engineer | GlideArc Solutions | Jul 2011 - Dec 2015

- Lowered field-reset incidents by 12% with watchdog recovery and diagnostics
- Validated 341 hardware test vectors across protocol and timing corner cases

PROJECTS

Bootloader Update Utility | Firmware Debugging, Oscilloscope, Embedded Linux, FreeRTOS

- Developed verified firmware transfer and rollback checks.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

Industrial Sensor Firmware | ARM Cortex-M, FreeRTOS, STM32, UART

- Built FreeRTOS tasks for calibrated sampling and watchdog recovery.
- Documented acceptance criteria and evaluated the design using hardware debugging and protocol validation.

EDUCATION

B.Tech in Electronics and Communication Engineering, Crescent Valley Technical University, 2011

Relevant Methods: Real-time systems, Hardware debugging, Protocol validation

CERTIFICATIONS

RTOS Scheduling Training (2022)